

Part II : Histology



Epithelial Tissue

Lect. 3

3) Glandular epithelium

GLANDS are cells or aggregations of cells whose function is secretion according to the presence or absence of their ducts can be divided into :

- **Endocrine glands** release their secretory product (typically hormones) into the spaces between the secretory cells (extracellular space) from which it enters the bloodstream.
- **Exocrine glands** release the secretory product via a system of ducts that opens upon one of the surfaces of the body which are in contact with the external world (skin, gastrointestinal tract etc.).

Classification of Exocrine Glands:

Exocrine glands may be classified according to cell number, and/or the shape and branching pattern of their secretory portions and ducts.

➤ according to cell number

■ Unicellular Glands :

consist of a single secretory cell. In mammals the only example of unicellular exocrine glands are ***goblet cells***, which occur in the epithelium of many mucous membranes. Goblet cells secrete the glycoprotein mucin, which by the uptake of water is converted into a slimy substance, mucus.

■ Multicellular glands .

1. According to presence or absence of ducts

According to presence or absence of ducts, the glands are classified into:

Endocrine

ductless glands secreting hormones directly in the blood as:

Thyroid, parathyroid, pituitary, suprarenal, pineal body, islets of Langerhans's, placenta, corpus luteum and special cells in the testis and ovary.

Exocrine glands

They have ducts to carry their secretions.
e.g., salivary glands, sweat and sebaceous glands.

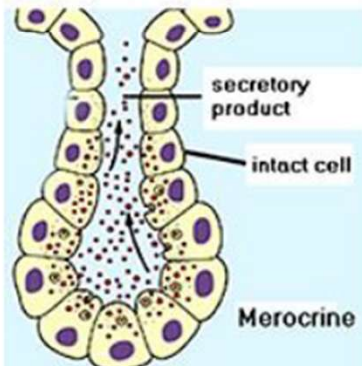
Mixed glands

which possess the exocrine and endocrine functions as:
pancreas, testis and ovary.

2. According to changes in the secretory cells

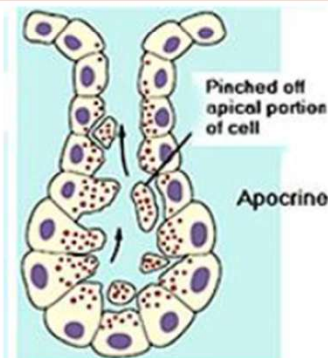
According to changes in the secretory cells, the glands are classified into:

Merocrine gland



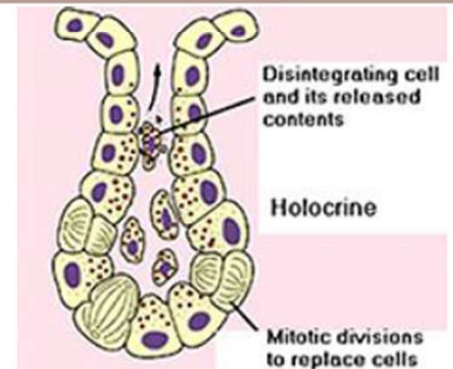
In these glands, there is no cellular changes in the secretory cells
ex: the salivary glands.

Apocrine gland



The tips or the secretory cell of the gland are detached and come out with the secretory products of the gland
ex: mammary glands and sweat glands.

Holocrine gland



The whole secretory cells are destroyed and come out with the secretion
ex: the cells of the sebaceous glands may come out with their secretion.

3. According to the kind of their secretion

The glands are classified according to the kind of their secretion into:

a) **Serous secretory glands:** as parotid gland and Von Ebner gland of the tongue.

b) **Mucous secretory glands:** as Brunner's gland and goblet cells..

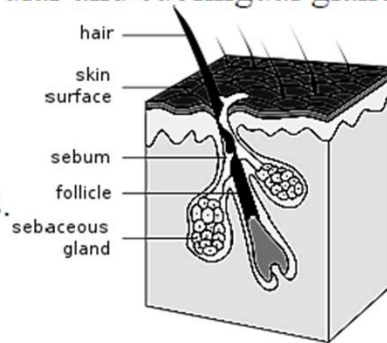
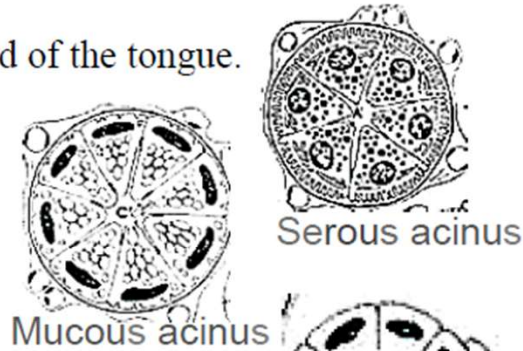
c) **Mucoserous secretory glands:** as submandibular and sublingual glands.

d) **Fatty secretory glands:** as sebaceous glands.

e) **Watery secretory glands:** as sweat glands.

f) **Waxy secretory glands:** as glands of external ear.

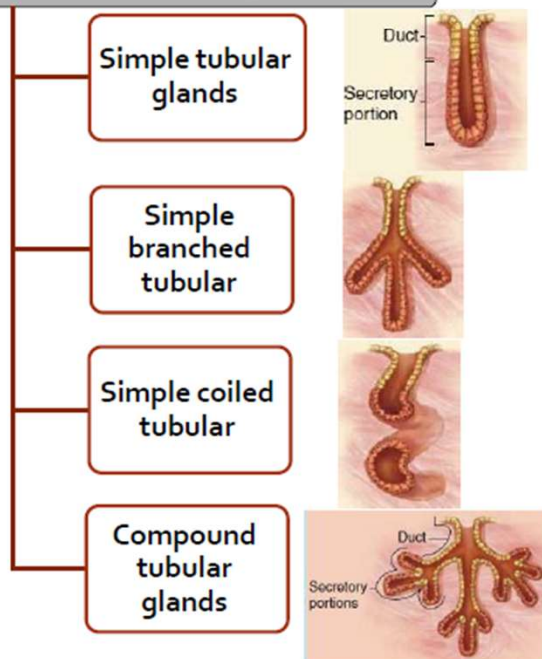
g) **Cellular secretory glands:** as testis and ovary.



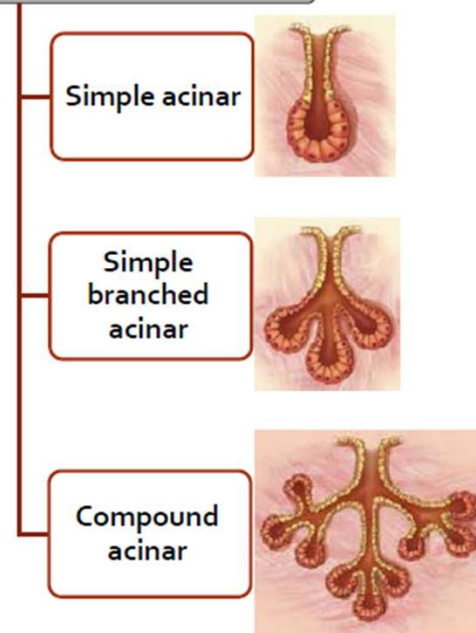
4. According to the shape and branching of the secretory part of the glands and the shape and branching of their duct

The shape of the glands may be Tubular, acinar or Tubulo-acinar.

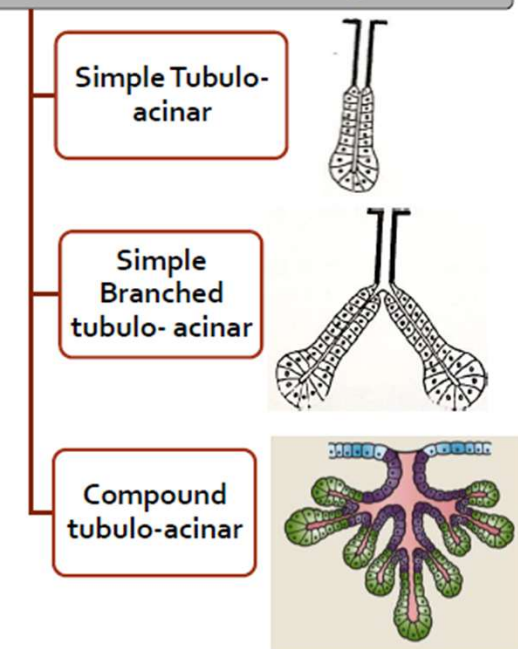
A - Tubular glands



B- Acinar glands



C- Tubulo-Acinar glands



5. According to their functions

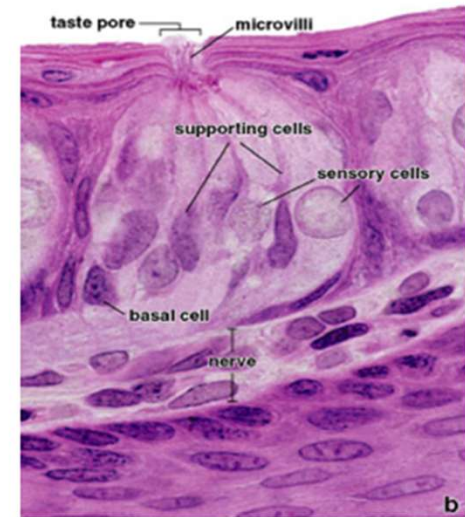
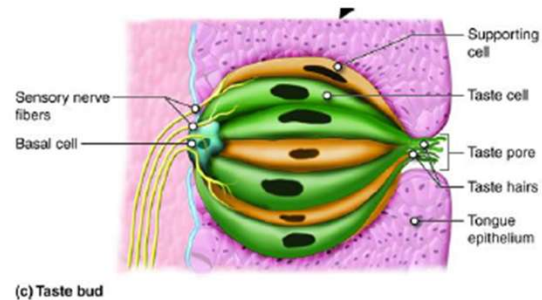
The glands can be classified according to their functions into: Secretory and Excretory Glands.

- a) **Secretory Glands**: which synthesize specific substances to be secreted in the body as salivary and endocrine glands.
- b) **Excretory Glands**: which eliminate and excrete the waste products outside the body as kidney and sweat glands.

D- Neuro-Epithelium

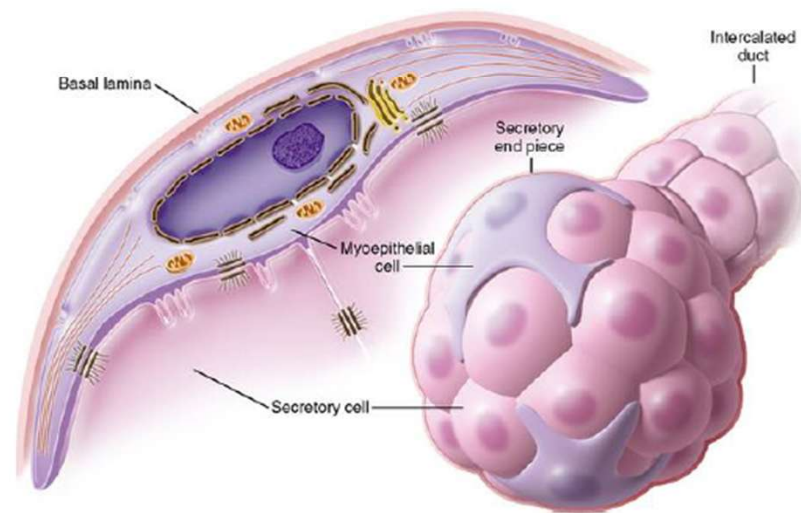
- It is the fourth type of epithelial tissue.
- In this type, the epithelial cells act as sensory receptors for special stimuli.
- The neuro - epithelial cells are provided with small hairs (hairlets) on their free surfaces, while their bases are surrounded with sensory nerves.

Neuro-epithelium is present in the taste buds of the tongue



E- Myo-Epithelium or Basket Cells

- Myoepithelial Cells or Basket cells are branched epithelial cells rich in actin and myosin filaments.
- They surround some secretory acini.
- They contract to squeeze the acini to secrete their products.
- They are present on the outer surfaces of the acini of sweat, salivary and mammary glands.



Functions Of Epithelial Tissues

1. **Protection** against injuries, bacteria, chemicals and water as epithelium of skin and stomach.
2. **Secretion** as glandular epithelium of pancreas, prostate, salivary and endocrine glands.
3. **Absorption** as the cells of intestine and kidney.
4. **Sensation** as the taste buds and organ of Corti.
5. **Reproduction** as the cells of the testis and ovary.
6. **Excretion** as the cells of kidney and sweat glands.
7. **Covering** surfaces (as skin) or **lining cavities** (as stomach).
8. **Respiration** as the epithelium of lung alveoli.